AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TT284
Web Technologies

Final Examination
Spring Semester 2025/2026

Date: 08/June/2026

AOU EGYPT

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 3 hours

Instructions:

- Total Marks: 100 **marks**
- This exam consists of **3 questions**
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Basic Calculators** are allowed.

Question 1: (True/False questions, 20 marks, 2 marks each one)

1. The most obvious thing about portable devices is that their screen size varies but is generally relatively big. As well as having fixed physical dimensions, they also have different screen size as measured in pixels.
2. Integration tests are intended to verify that individual units work together, which for the typical web application means testing functionality, usability, browser compatibility, and accessibility.
3. Mobile smartphone and tablet manufacturers do not offer any Software Development Kits (SDKs), making native app development impossible without third-party tools.
4. Authentication relates to permissions; it's about what I am permitted to do, which might be related to who I am, but not necessarily so.
5. Risk management is a field of study that brings together a wide range of techniques and tools for managing any development activity. It can help to organise the tasks and manage the resources so that an application gets completed on time, on budget, and most importantly, fulfils its requirements.
6. GPS signals are difficult to receive inside a building and obtaining a lock on the satellites can take several minutes, particularly if the signal from them is weak.
7. No matter how good the installation and configuration process, errors are made and applications are still subject to attacks that attempt to exploit new vulnerabilities. It is therefore essential to record all the activities that take place on a server so that we can assess if anyone is attempting to attack our application.
8. One of the more common metrics associated with reliability is the 'mean time to failure' (MTTF), which is a statistical measure of reliability for items that cannot be repaired. It is based on testing a batch of items over a short period of time, where 'short' is taken to mean short when compared to the expected life of the item, perhaps two months of testing for a product with a 5-year lifespan.
9. XHTML validation, and CSS validation are some of the items that should be covered by the unit test for a typical web application.
10. Environmental factors, Application errors, and Operator and user errors are some of the causes of downtime of the systems.

Question 2: (Essay questions, 30 marks, 5 marks each one)

a. List five of the potential benefits of virtualisation.
b. Discuss four of the stages of the Waterfall lifecycle model.
c. Discuss the Load test of the load performance testing, show an example.
d. The Consortium for IT Software Quality has defined the structural quality attributes essential for a software to deliver business value. Discuss these attributes.
e. Explain briefly the new features of the V-model.
f. Discuss four of the key sections of a generic test plan.

Question 3: (Problem Solving Questions, 50 marks, 10 marks each one)

a. A disk drive manufacturer tested a number of disk drives several times, at the end of the testing period, two disk drives were found to have failed; If the mean time to failure = 200000 hours, calculate the annualised failure rate.

Not Yet

b. Write a JavaScript program that evaluates a website's security level based on the following criteria in Table 1:

Criteria	Secure Value input	Score
Uses HTTPS	Y/N	If input = "Y" then Score =1 else score = 0
Has a valid SSL certificate	Y/N	If input = "Y" then Score =1 else score = 0
Has security headers enabled	Y/N	If input = "Y" then Score =1 else score = 0
Number of known vulnerabilities	Number	If input <=1 then Score =1 else score = 0
Last security audit (in months)	Number	If input <=3 then Score =1 else score = 0

Table 1

The output will be based on the following Classification:

1. Secure: All criteria met
2. Needs Improvement: 3 to 4 out of 5 criteria met
3. Insecure: 0 to 2 out of 5 criteria met

c. Write HTML code that will create the table shown in Figure 1.

Students Grades

Book	Author	Price	Available?
HTML	John	$30	Yes
JavaScript	Huda	$25	Yes
CSS	Diana	$20	No

Figure 1

d. What is the output of the following program?

```
<html>    <body>    <script>
    var NUM = -6;
    var ST = 3;
    var ED = 9;
    for (i = 0; i < 10; i = i + 2)  {
      document.write(NUM + "<br>");
      if (ST >= 3 && ED <10)  {
         NUM = NUM + 2;
         ST = ST - 2;
         ED = ED - 2;
       }
      else
        if (ED <= 8) {
          NUM = NUM + 2;
```

ACCEPT

```
            ST = ST + 2;
            ED = ED + 2;
          }
        }
</script>    </body>    </html>
```

e. Develop a simple web page containing the following specification:

1. Page Title “Flight Ticket”.
2. Header 1 “Flight Ticket Sales”.
3. Form action to locate the script that will process the form data “http//norton.open.ac.uk/reflect.php”.
4. Image that displays the logo saved in the Logo.GIF file.
5. Text box for the First Name.
6. Text box for the Last Name.
7. Text box for the e-mail.
8. Username text field.
9. Password field.
10. Upload identification card file.
11. Fieldset1 Radio button for the First Class.
12. Fieldset1 Radio button for the Special Class.
13. Fieldset1 Radio button for the Economy Class.
14. Fieldset2 Radio button for the Window Seat.
15. Fieldset2 Radio button for the Middle Seat.
16. Fieldset3 Checkbox for the Meal Services.
17. Fieldset3 Checkbox for the Hosting Services.
18. Fieldset3 Checkbox for the Extra Weight Services.
19. Submit button with submit value.
20. Reset button with Reset value.

The web page should look like the following design (Figure 2):



Figure 2

End of questions